E-commerce Checkout Funnel Analysis

1. **Objective**

The objective of this project was to analyze the end-to-end user journey in an e-commerce platform and identify key drop-off points impacting conversion rate and revenue. The goal was to provide actionable insights to improve funnel efficiency and overall business performance.

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2. **Business Context**

In e-commerce, a significant portion of users do not complete the purchase journey, leading to revenue loss. Understanding where and why users drop off in the funnel is critical to improving conversion rates without increasing customer acquisition costs.

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3. **Data Overview**

The analysis was based on event-level user data capturing key stages of the funnel:

- Product View
- Add to Cart
- Checkout
- Purchase

Additional attributes included:

- User ID
- Device type (Desktop, Mobile, Tablet)
- Country
- Transaction ID
- Revenue

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4. **Tools & Technologies**

- SQL → Data extraction and transformation
- Excel → Data validation and intermediate analysis
- Power BI → Data modelling and dashboard development

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5. **Approach**

- Mapped user events to defined funnel stages
- Created stage-wise user counts to build funnel flow
- Calculated conversion rates and drop-off percentages between stages
- Performed segmentation analysis by device and geography
- Analyzed revenue metrics including total revenue, AOV, and revenue loss

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6. **Key Insights**

- Approximately **87% of users drop before adding items to cart**, indicating a major issue at the product page level
- **Cart abandonment rate is ~62%**, suggesting friction in the checkout process
- Overall **conversion rate is ~2.83%**, significantly below expected benchmarks
- **Mobile users show lower conversion compared to desktop**, indicating UX or performance issues
- Estimated **revenue loss is high due to funnel inefficiencies**, especially at early and mid-stages

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7. **Business Impact**

The analysis highlights that improving conversion at critical funnel stages can significantly increase revenue without additional marketing spend. Even a small improvement in cart or checkout conversion can lead to measurable revenue growth.

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8. **Recommendations**

- Improve product page experience (clear CTA, pricing transparency, product details)
- Simplify checkout process (reduce steps, enable guest checkout, auto-fill forms)
- Optimize mobile user experience (speed, UI responsiveness)
- Implement retargeting strategies for cart abandoners
- Introduce trust elements (reviews, secure payment indicators)